5 **(a)** **(i)** State what is meant by *a line of force in a gravitational field*.

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……[1]

(ii) By reference to the lines of force of the gravitational field near the surface of the Earth, explain why the acceleration of free fall near the surface of Earth is approximately constant.

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……[3]

(b) The table below shows some data for Mercury and Pluto.

	Mass/kg	Radius/m	Mean distance from Sun/m
Mercury	3.30×10^{23}	2.44×10^{6}	57.9×10^{9}
Pluto	0.131×10^{23}	1.19×10^{6}	5910×10^{9}

(i) Show that the escape velocity v of a gas molecule on the surface of Pluto is given by the equation

$$v = \sqrt{\frac{2GM}{r}}$$

where M is the mass of Pluto and r is its radius. [2]

(ii) Calculate the escape velocity of a gas molecule on the surface of Pluto.

escape velocity = ……m s^{-1} [1]

(iii) Suggest, using data from the table, why Mercury has no detectable atmosphere while Pluto has a thin atmosphere.

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……[3]